Day 4: Continuous Integration (CI) Concepts Assignment

Company Overview
Motion Recruitment is a leading staffing agency specializing in connecting top tech talent with innovative companies. With years of experience in the technology recruitment space, Motion Recruitment focuses on understanding both client and candidate needs to create successful matches. Our commitment to quality service and deep industry expertise enables us to deliver tailored solutions for various technology roles, ensuring that businesses have the right talent to drive their success.

Task Information

As a new member of the DevOps team at Motion Recruitment, you are tasked with demonstrating your understanding of Continuous Integration (CI) concepts and setting up a CI pipeline across AWS, Azure, and GCP.

Assignment Details

Objective

Demonstrate your understanding of CI principles, benefits, and the ability to set up a basic CI pipeline on all three cloud platforms.

Requirements

1. **Continuous Integration Principles**
   a. Write a brief document explaining the principles of Continuous Integration and its benefits for software development. Include examples of how CI can improve collaboration and code quality.
2. **Setting Up a Basic CI Pipeline on AWS**
   a. Create a new branch named **"feature-aws-ci."**
   b. Add a file named **"aws-ci-setup.md"** to this branch. Inside the file, document the steps required to set up a CI pipeline using **Jenkins** on AWS.
   c. Define pipeline stages:
   - **Build:** Describe how to configure Jenkins to build your application (or sample Python file).
   - **Test:** Detail how to run tests as part of the CI process.
   - **Deploy:** Explain how to deploy the application to an EC2 instance.
     d. Configure build triggers to automatically start builds on code changes.
     e. Commit your changes with an appropriate message.
     f. Push your changes and create a pull request to merge **"feature-aws-ci"** into the main branch.
3. **Setting Up a Basic CI Pipeline on Azure**
   a. Create a new branch named **"feature-azure-ci."**
   b. Add a file named **"azure-ci-setup.md"** to this branch. Inside the file, document the steps required to set up a CI pipeline using **GitLab CI** or **Azure DevOps** on Azure.
   c. Define pipeline stages:
   - **Build:** Describe how to configure the build process.
   - **Test:** Detail how to run tests during the CI process.

- **Deploy:** Explain how to deploy the application to an Azure VM.
  d. Configure build triggers to automatically start builds on code changes.
  e. Commit your changes with a descriptive message.
  f. Push your changes and create a pull request to merge **"feature-azure-ci"** into the main branch.

4. **Setting Up a Basic CI Pipeline on GCP**
   a. Create a new branch named **"feature-gcp-ci."**
   b. Add a file named **"gcp-ci-setup.md"** to this branch. Inside the file, document the steps required to set up a CI pipeline using **CircleCI** on GCP.
   c. Define pipeline stages:
   - **Build:** Describe how to configure the CircleCI build process.
   - **Test:** Detail how to run tests as part of the CI process.
   - **Deploy:** Explain how to deploy the application to a Google Compute Engine instance.
     d. Configure build triggers to automatically start builds on code changes.
     e. Commit your changes with a clear message.
     f. Push your changes and create a pull request to merge **"feature-gcp-ci"** into the main branch.
5. **Final Review and Integration**
   a. After merging all three CI branches (AWS, Azure, and GCP), ensure that the main branch contains documentation for all three CI setups in the corresponding markdown files.
   b. Create a new branch named **"feature-final-ci-review."**
   c. Review and finalize all documentation, ensuring clarity and completeness.
   d. Merge the final review branch back into the main branch.

Implementation Guidelines

- Use clear and descriptive commit messages throughout your process.
- Follow best practices for CI pipeline configuration.
- Ensure that each cloud CI setup is documented clearly in its respective markdown file.

Submission Instructions

1. Ensure all the changes are pushed to your GitHub repository.
2. Add the collaborator's review and your responses to each pull request conversation.
3. Provide the link to your GitHub repository for evaluation.

Evaluation Criteria

Your assignment will be evaluated based on:

1. **Correct implementation of CI principles and setups** (Jenkins, GitLab CI, CircleCI).
2. **Quality of documentation** for AWS, Azure, and GCP CI pipelines.
3. **Effective collaboration using GitHub** (pull requests, code reviews, addressing feedback).
4. **Clear and descriptive commit messages.**
5. **Overall repository structure and organization.**

Good luck with your assignment!